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GeoCT-SigNet: A Geometry-Aware Hybrid CNN–Transformer Network for Offline Signature Verification

HAIDER ALI[1], MOMINA MOETESUM[1], ADNAN UL-HASAN[1,2], FAISAL SHAFAIT[1,2], and IMRAN MALIK[1,2]

[1]School of Electrical Engineering and Computer Science (SEECS), National University of Sciences & Technology (NUST), Islamabad, Pakistan (e-mail: hali.msai22seecs@seecs.edu.pk; momina.moetesum@seecs.edu.pk; adnan.ulhassan@seecs.edu.pk; faisal.shafait@seecs.edu.pk; malik.imran@seecs.edu.pk)
[2]Deep Learning Laboratory, National Center of Artificial Intelligence (NCAI), Islamabad, Pakistan

Corresponding author: Imran Malik (e-mail: malik.imran@seecs.edu.pk).

ABSTRACT Offline handwritten signature verification (OHSV) requires simultaneous modeling of fine stroke-level geometry and high-level semantic structure, yet neither convolutional nor transformer-based architectures achieve this independently. Recent hybrid CNN–Transformer models improve local-global integration but remain insensitive to high-frequency contour details critical for distinguishing genuine signatures from skilled forgeries. We propose GeoCT-SigNet, which addresses this gap by embedding the contourlet transform (a multi-scale, multi-directional image decomposition) into a differentiable Siamese pipeline. A Frequency-Aware Feature Extraction (FAFE) block refines low- and high-frequency subbands independently, fusing them with deep semantic features via bidirectional cross-attention and a depth-wise feed-forward module. GeoCT-SigNet achieves state-of-the-art (SOTA) EER on BHSig-B (5.80%), BHSig-H (6.84%), and competitive performance on UTSig (18.99%). At the real-forgery operating threshold, it attains 0% FAR against CGAN-generated forgeries and $\leq 3.6\%$ against VAE-generated forgeries, substantially outperforming CNN and transformer baselines. At only 4.13M parameters, it is $11\times$ more compact than the closest hybrid baseline, establishing contourlet-based geometric priors as a promising direction for parameter-efficient, robust OHSV.

INDEX TERMS Contourlet transform, deep learning, offline signature verification, signature forgery detection, siamese network, synthetic signatures

I. INTRODUCTION

HANDWRITTEN signatures have served as a primary means of personal authentication for centuries, and despite the rapid digitization of legal, financial, and administrative processes, they remain legally binding identifiers across banking, contract execution, and identity verification worldwide [1], [2]. This continued reliance has made automated Offline Handwritten Signature Verification (OHSV) an increasingly critical component of document security infrastructure [3]. Unlike online systems that record temporal signals such as pressure and velocity, offline systems must extract all discriminative information from a single two-dimensional image, making the verification task fundamentally more challenging and the need for robust feature representation more acute [4].

OHSV systems can be categorized as writer-dependent or writer-independent. Writer-dependent systems, tailored for individual writers, require large sample sets and retraining for each new writer, which limits scalability, whereas writer-independent systems focus on distinguishing genuine from forged samples and adapt quickly to new writers [3]. These systems face challenges due to intrapersonal variability (i.e. differences among signatures from the same writer) and interpersonal variability (i.e. differences between writers) increased by various forgery types, including random forgeries (using an imposter's own signature), simple forgeries (attempted reproductions based solely on the signer's name), and expert forgeries (carefully replicated victim signatures), with expert forgeries presenting the most critical test for verification [5].

Traditional methods for signature verification rely on handcrafted feature extractors based on manually constructed prior knowledge [6]. However, these methods rely on prior assumptions which limits their generalizability and scalability. In comparison, deep learning methods such as Convolutional Neural Network (CNN), have demonstrated better performance and require fewer prior assumptions about the data [7]–[10]. However, CNN-based models require deep networks to learn global features of images, and convolution operations are inherently limited in modeling long-range dependencies [11]. To overcome this, various studies [12]–[14] have proposed transformer-based OHSV systems that achieve superior performance in signature verification. However, transformers tend to dilute local stroke details while attempting to extract high-level semantic features [15]. To overcome this issue, recent methods [16], [17] proposed hybrid models that combine capabilities of CNN and Transformer models for the signature verification task. However, these methods are less effective in capturing high-frequency, detailed representations (such as edges, strokes, and contours) [18]. Because they still rely on broad semantic features not geometric micro-structure. Consequently, this limitation of these models to capture micro-structure geometric details not only limits their performance against expert forgeries but can also increase their vulnerability to emerging synthetic forgeries. Recent studies [19], [20] have highlighted this new verification challenge for OHSV systems. These studies used Conditional Generative Adversarial Network (CGAN) [21] and Variational Auto Encoder (VAE) [22] to generate synthetic forgeries to expose vulnerabilities of existing OHSV systems.

In this article, we propose a novel geometry aware hybrid CNN-Transformer-based signature verification network (GeoCT-SigNet). GeoCT-SigNet performs multi-scale and multi-directional feature extraction on the signature images, allowing the network to capture deep detail feature representation. Specifically,within a Geometric Feature Extraction (GFE) branch, a dedicated Frequency-Aware Feature (FAFE) sub-block independently refines both the low-frequency (structural) and high-frequency (contour) contourlet subbands through parallel convolutional branches and reconstructs a geometrically enhanced signature image through the Inverse Contourlet Transform (ICT); this refined image is then consumed by a deep CNN feature extraction (DCNNFE) module to learn local structural features. Additionally, it applies Deep Semantic Feature extraction (DSFE) to capture high-level semantic features. Subsequently, it applies the Cross-Feature Refinement (CFR) module to capture cross-dependency relationships between high-level semantic features and the refined geometric feature representations produced by FAFE. These features are fused through the Depth-wise Feed-Forward Fusion (DWFFF) module to produce the final feature embedding for verification.

The main contributions of the paper are outlined below:

1) We have proposed a novel end-to-end Geometry Aware Hybrid CNN–Transformer Signature Network (GeoCT-SigNet), which effectively integrates geometric micro-structure and deep semantic features of offline signatures into network optimization, producing superior performance in the writer-independent signature verification task.
2) Our proposed geometric feature extraction block effectively captures the multi-scale structural and multi-directional contour representation of the signature image, which is then integrated with deep semantic features by a depth-wise feed-forward fusion block for a rich feature representation.
3) We conduct extensive comparative experiments and ablation studies, showing that GeoCT-SigNet outperforms state-of-the-art signature verification systems across four benchmark datasets on the signature verification task. Moreover, we have also evaluated GeoCT-SigNet on synthetic signatures generated by two widely used generative models (i.e. CGAN and VAE), showing its robustness against such synthetic signatures.

The structure of the paper is as follows. Section II discusses the related works. Section III details the methodology of the proposed approach. Section IV describes the experimental protocol. Section V presents experimental results and analysis and Section VI concludes the paper.

II. RELATED WORK

Offline Handwritten Signature Verification (OHSV) can be considered a two-class problem where the target is to decide whether a signature is genuine or forged [3]. Earlier research by Kalera et al. [23] used distance metrics to differentiate between genuine and forged signatures, achieving as high as 93% identification of writer and 78% verification of genuine and forged signatures on two distinct datasets. Dominique et al. [24] proposed a method for extracting multiple features, enabling the extraction of global and local features by adjusting the scale of a virtual grid that divides each signature image. Hamadene et al. [4] proposed a system that involves the use of a contourlet transform-based directional code co-occurrence matrix feature generation method. Although Hamadene et al. [4] also exploit the directional selectivity of the contourlet transform for OHSV, their pipeline applies the transform as a fixed front-end whose coefficients are summarised by hand-crafted directional code co-occurrence matrices and fed to a non-trainable feature-dissimilarity classifier. Our approach differs along three concrete axes. First, the contourlet decomposition in GeoCT-SigNet is wrapped inside a fully differentiable computation graph (FAFE $\rightarrow$ ICT) so that the per-band convolutional

refinements, the inverse-contourlet reconstruction, and the downstream CNN/Transformer branches are jointly optimised end-to-end through contrastive backprop; no hand-crafted statistics gate the gradient flow. Second, the low-frequency and high-frequency contourlet subbands are refined by independent learnable convolution blocks rather than being collapsed into a single co-occurrence descriptor, which lets the network adapt the relative contribution of structural and contour cues per dataset and per script. Third, the geometric stream is fused with deep semantic features (DSFE) through bi-directional cross-attention (CFR) and a depth-wise feed-forward fusion block (DWFFF), so the contourlet representation participates in the final embedding rather than serving as a stand-alone classifier. The net effect is that GeoCT-SigNet inherits the multi-resolution / multi-direction inductive bias of the contourlet transform while operating in a writer-independent, end-to-end-learnable regime that is not available to the handcrafted-feature pipeline of Hamadene et al.

Due to difficulty in feature extraction, recent studies' focus has been on learning feature representation of the signature images rather than relying on handcrafted feature extractors. Feature representations are learned by deep learning techniques such as Convolutional Neural Networks (CNNs) [3]. In [7], Sounak et al. proposed a convolutional siamese network (SigNet) for writer-independent offline signature verification. This approach utilizes twin convolutional neural networks (CNNs) to learn a feature space that captures the intrinsic properties of individual signatures and classifies genuine signatures based on a similarity metric. The study formulated the signature verification task as a few-shot learning problem that inspired future works [25], [26] in this direction.

Although CNNs have achieved remarkable performance in the OHSV system, their reliance on local neighborhoods makes it challenging to capture global semantic relationships among signature strokes [11]. To overcome this, recent studies have integrated attention mechanisms and transformer-based architectures to model long-range dependencies and effectively focus on discriminative stroke regions. Wei et al. [27] introduced an Inverse Discriminative Network (IDN) employing dual attention pathways between original and inverse images to emphasize relevant signature features. Inspired by this, Xiong et al. [26] proposed an Attention-based Multiple Siamese Network (MSN) to enhance feature comparison across paired signatures. Transformer-based architectures, such as Li et al.'s TransOSV [13] and Ren et al.'s 2C2S [14], leverage self-attention to jointly encode global and local feature representations, thereby improving robustness against skilled forgeries. As transformer-based models dilute low-level features, Zheng et al. [16] proposed HTCSigNet, which combines CNNs with Vision Transformers to balance fine-grained local detail and holistic context, achieving superior generalization across standard datasets.

Although Transformer–CNN hybrid approaches have shown promise in balancing global and local feature representations, they are still confronted with a contradiction between comprehensive semantic understanding and meticulous detail extraction [18]. To address this, we have proposed GeoCT-SigNet. GeoCT-SigNet is a novel end-to-end signature verification system. It integrates low-level structural features and high-level semantic features with multi-scale and multi-directional geometric features to capture a meticulous and detailed representation of the signature image.

III. PROPOSED METHODOLOGY

We explain our proposed approach in detail in this section. We will begin by discussing the proposed Writer Independent Offline Handwritten Signature Verification (OHSV) system, as illustrated in Fig. 1. Afterward, we will discuss the feature generation method (i.e., GeoCT-SigNet).

A. WRITER-INDEPENDENT OHSV

Our proposed OHSV system is based on a Siamese network architecture consisting of twin subnetworks with shared weights as shown in Fig. 1. These networks learn a feature space where similar samples are placed closer together by minimizing the distance between positive pairs (i.e., genuine–genuine) and maximizing it between negative pairs (i.e., genuine–forged). Training involves both positive and negative signature pairs using the contrastive loss:

$$\begin{aligned} \mathrm{L}(\mathrm{p}_1, \mathrm{p}_2, \mathrm{y}) &= \mathrm{y}\,\mathrm{D}_\mathrm{w}^2 + (1-\mathrm{y})\max(0, \mathrm{m} - \mathrm{D}_\mathrm{w})^2, \\ &\text{where } \mathrm{p}_1, \mathrm{p}_2 \in \mathrm{X}. \end{aligned} \tag{1}$$

Here, p_1 and p_2 denote two input signatures, and $\mathrm{y} \in \{0, 1\}$ indicates whether the pair belongs to the same writer, with $\mathrm{y} = 1$ for genuine–genuine (positive) pairs and $\mathrm{y} = 0$ for genuine–forged (negative) pairs. Under this convention, the first term $\mathrm{y}\,\mathrm{D}_\mathrm{w}^2$ pulls positive pairs together, while the second term $(1-\mathrm{y})\max(0, \mathrm{m} - \mathrm{D}_\mathrm{w})^2$ pushes negative pairs apart whenever their distance falls below the margin. The two loss terms are weighted symmetrically, following the standard contrastive loss formulation [28]. m is the margin (set to 1). The Euclidean distance D_w is given by:

$$\mathrm{D}_\mathrm{w} = \|\mathrm{f}(\mathrm{s}_1; \mathrm{w}_1) - \mathrm{f}(\mathrm{s}_2; \mathrm{w}_2)\|_2, \tag{2}$$

where f is the embedding function that maps signature pixels into a real-valued vector space through our proposed feature extraction. The network learns weights w_1 and w_2 at each layer. The input comprises signature pairs: positive pairs and negative pairs. Positive pairs are encouraged to produce smaller distances than negative pairs.

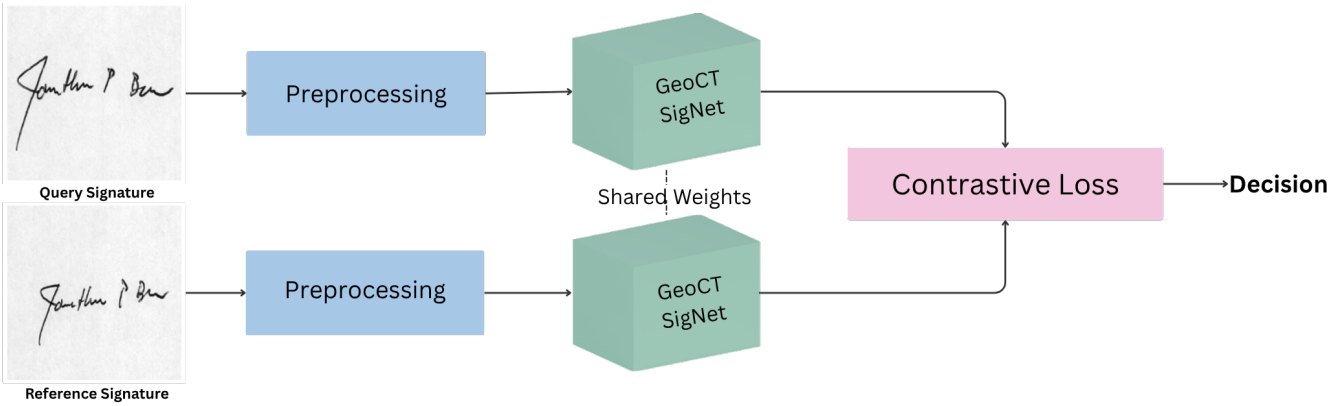


FIGURE 1. Overview of the proposed Siamese GeoCT-SigNet architecture for offline signature verification. The query and reference signatures are independently preprocessed and forwarded through the shared GeoCT-SigNet to generate embedding vectors. These embeddings are compared using a contrastive loss function to produce the final verification decision.

B. GEOCT-SIGNET OVERVIEW

Our proposed GeoCT-SigNet (Fig. 3) consists of three feature extraction modules: Geometric Feature Extraction (GFE), Deep CNN Feature Extraction (DCNNFE), and Deep Semantic Feature Extraction (DSFE). GFE uses a multi-scale geometric contourlet transform to capture multi-directional contours and fine structural details. Inside GFE, a Frequency-Aware Feature (FAFE) sub-block independently refines the low-frequency and high-frequency contourlet subbands through parallel Conv→IN→ReLU branches; the refined bands are (i) recombined by the Inverse Contourlet Transform (ICT) to produce a geometrically enhanced image for DCNNFE, and (ii) fused in feature space by a concatenation followed by a convolution and an MLP projection to produce the geometric descriptor F_{lh}; it also computes a compact Contourlet Statistical Feature vector (F_{SCT}) by collecting the mean and standard deviation of each subband coefficient map, providing lightweight handcrafted descriptors complementary to the learned representations. DCNNFE learns local features from the geometrically refined signature reconstructed by FAFE, while DSFE, based on a Transformer, captures high-level semantic dependencies. A Cross-Feature Refinement (CFR) block then fuses the GFE geometric output F_{lh} and the DSFE semantic output F_{DS} via bi-directional cross-attention, producing a refined feature F_{SF}. These feature representations (F_{SF}, F_{DF}, F_{SCT}) are fused via a Depth-Wise Feed-Forward Fusion (DWFFF) block to produce a unified feature embedding optimized with contrastive loss.

Throughout this section, $X_{img} \in \mathbb{R}^{H\times W\times C}$ denotes a raw signature image and X'_{img} denotes its preprocessed counterpart (resized to 224×224, contrast-inverted, and pixel-normalised as described in Section IV).

C. MULTI-SCALE GEOMETRIC FEATURE EXTRACTION

The Geometric Feature Extraction (GFE) applies the Contourlet Transform (CT), introduced by Do and Vetterli [29], to extract multi-scale and multi-directional features from the input image. CT extends the wavelet transform [30] through multi-scale, directional decomposition, effectively capturing smooth contours and edges in signature images. It first uses a Laplacian Pyramid (LP) [31] to decompose the image into low-frequency (capturing overall structure) and high-frequency (isolating edges and textures) subbands. The LP for a preprocessed signature image X'_{img} can be mathematically represented as:

$$\begin{cases} X^1_{low} = (G * X'_{img})\downarrow, \\ X^1_{high} = X'_{img} - \mathcal{H}\,(X^1_{low}\uparrow) \end{cases} \tag{3}$$

where $\mathcal{H}$ represents the reconstruction (synthesis) filter and $\mathcal{G}$ represents the analysis filter, and $*$ denotes convolution. $\uparrow$ and $\downarrow$ represent upsampling and downsampling, respectively. Subsequently, the high-frequency subband X^1_{high} is passed through a directional filter bank (DFB) [32], splitting them into 2^d directional subbands (d = decomposition level). By iteratively applying these steps to the low-frequency subband X^1_{low}, CT yields a multi-scale, multi-directional representation that effectively captures high-frequency detail representation of handwritten signatures as shown in Fig. 2.

Given a preprocessed signature image $X'_{img} \in \mathbb{R}^{H\times W\times C}$ where H represents height, W represents width, and C represents channels, we apply CT at level l to decompose it into a low-frequency subband (LP) and high-frequency (directional) subbands. Both frequency bands carry complementary and equally important information for signature verification: the low-frequency subband encodes coarse structural and semantic content, whereas the high-frequency subbands isolate fine stroke edges that are most discriminative of a writer's identity.

Accordingly, we refine each band independently with its own convolution block to learn scale-specific features. Furthermore, we apply instance normalization and ReLU for a more stable gradient flow. Mathematically, it can be represented as:

$$\begin{cases} \mathrm{X}^{\mathrm{l}}_{\mathrm{low}},\ \mathrm{X}^{\mathrm{l}}_{\mathrm{high}} = \mathrm{CT}\left(\mathrm{X}'_{\mathrm{img}}\right), \\ \mathrm{X'}^{\mathrm{l}}_{\mathrm{low}} = \mathrm{R}\left(\mathrm{X}^{\mathrm{l}}_{\mathrm{low}}\right), \\ \mathrm{X'}^{\mathrm{l}}_{\mathrm{high}} = \mathrm{R}\left(\mathrm{X}^{\mathrm{l}}_{\mathrm{high}}\right) \end{cases} \tag{4}$$

where CT represents the Contourlet Transform, l represents the l-th level of LP, and R(·) represents Conv+InstanceNorm+ReLU. Together, the two refinement branches constitute the Frequency-Aware Feature (FAFE) block referenced in Table 1. Because the two bands differ in spatial resolution, channel count, and statistical distribution, each branch uses its own learnable parameters rather than a shared set of weights; this allows the network to tailor its response to the coarse-scale structural cues on one hand and to the sparse, anisotropic contour responses on the other.

The high-frequency tensor $\mathrm{X}^{\mathrm{l}}_{\mathrm{high}}$ already contains the 2^{d} directional subbands produced by the directional filter bank [32] that follows the Laplacian Pyramid inside the Contourlet Transform, where d denotes the DFB decomposition depth. Refining these bands with R(·) therefore performs a direction-selective denoising that attenuates isotropic noise while preserving anisotropic stroke structures such as contours.

In parallel to the ICT reconstruction pathway described next, we concatenate feature maps extracted from the FAFE-refined low- and high-frequency bands and pass them through another convolution block R(·) to learn a structural and detailed geometric representation. Mathematically, it can be represented as:

$$\mathrm{G}_{\mathrm{lh}} = \mathrm{R}\left(\left[\mathrm{X'}^{\mathrm{l}}_{\mathrm{low}},\ \mathrm{X'}^{\mathrm{l}}_{\mathrm{high}}\right]\right) \tag{5}$$

where G_{lh} represents the feature space of geometric representation of the input signature image. The resulting feature map is then reduced to a compact vector representation through adaptive global average pooling followed by a linear projection (realised by an MLP layer in Fig. 3):

$$\mathrm{F}_{\mathrm{lh}} = \mathrm{W}_{\mathrm{p}}\,\mathrm{GAP}(\mathrm{G}_{\mathrm{lh}}) + \mathrm{b}_{\mathrm{p}}, \tag{6}$$

where GAP(·) denotes adaptive average pooling to spatial size 1×1, and $\mathrm{W}_{\mathrm{p}} \in \mathbb{R}^{\mathrm{d}\times\mathrm{c}}$ projects the c-channel pooled descriptor to the d-dimensional embedding space shared across all subsequent modules. Global average pooling is chosen because it removes spatial dependency and yields a translation-invariant descriptor, which is desirable given the slight positional variance of handwritten signatures. The resulting vector F_{lh} is the geometric-stream input to the Cross-Feature Refinement (CFR) block.

Subsequently, we reconstruct a geometrically refined signature image using the FAFE-refined low- and high-frequency subbands $\mathrm{X'}^{\mathrm{l}}_{\mathrm{low}}$ and $\mathrm{X'}^{\mathrm{l}}_{\mathrm{high}}$ using the inverse contourlet transform. Mathematically, it can be represented as:

$$\mathrm{X}'_{\mathrm{rec}} = \mathrm{ICT}\left(\mathrm{X'}^{\mathrm{l}}_{\mathrm{low}},\ \mathrm{X'}^{\mathrm{l}}_{\mathrm{high}}\right) \tag{7}$$

Reconstructing from both refined bands ensures that coarse structural context and fine contour detail are simultaneously sharpened before being passed to the CNN backbone: the low-frequency branch enhances inter-stroke spatial relations that are often blurred by image noise, while the high-frequency branch emphasises the anisotropic stroke boundaries that carry most of the writer-specific micro-structure. Because both refinements operate on the critically sampled contourlet coefficients and the synthesis filters are the adjoints of the analysis filters, the ICT step still respects the perfect-reconstruction property of the contourlet filter banks [29], so any residual distortion is confined to what the two convolution branches have learned to emphasise. The reconstructed image $\mathrm{X}'_{\mathrm{rec}}$ is fed into a DCNNFE module to extract local structural features.

1) Contourlet Statistical Features

Given a preprocessed input signature image $\mathrm{X}'_{\mathrm{img}}$, the CT decomposes an input image into multi-scale and multi-directional coefficients. These coefficients contain high-level information of the input image $\mathrm{X}'_{\mathrm{img}}$. These coefficients' mean and standard deviation can represent useful information on smooth contours in a handwriting signature. The index of coefficients, the mean μ_{n} and standard deviation σ_{n} are chosen as the texture features of decomposed images. Thus, the contourlet statistical feature $\mathrm{F}_{\mathrm{SCT}}$ is formulated as:

$$\begin{aligned} \mathrm{F}_{\mathrm{SCT}} &= \{\mu_1, \sigma_1, \mu_2, \sigma_2, \ldots, \mu_{2^{\mathrm{l}}}, \sigma_{2^{\mathrm{l}}}\} \\ \mu_{\mathrm{n}} &= \frac{1}{\mathrm{W}\times\mathrm{H}} \sum_{\mathrm{x}=1}^{\mathrm{W}} \sum_{\mathrm{y}=1}^{\mathrm{H}} |\mathrm{C}_{\mathrm{n}}(\mathrm{x},\mathrm{y})| \\ \sigma_{\mathrm{n}} &= \sqrt{\frac{1}{\mathrm{W}\times\mathrm{H}} \sum_{\mathrm{x}=1}^{\mathrm{W}} \sum_{\mathrm{y}=1}^{\mathrm{H}} \left(|\mathrm{C}_{\mathrm{n}}(\mathrm{x},\mathrm{y})| - \mu_{\mathrm{n}}\right)^2} \end{aligned} \tag{8}$$

where $\mathrm{l} = 1, 2, \ldots, \mathrm{L}$ represents the number of decomposition levels, $\mathrm{C}_{\mathrm{n}}(\mathrm{x},\mathrm{y})$ is the nth contourlet subband coefficient of an image, and the width and height of the nth subband are W and H, respectively.

D. DEEP CNN FEATURE EXTRACTION

After Geometric Feature Extraction, the reconstructed image $\mathrm{X}'_{\mathrm{rec}}$ is passed through the DCNNFE module to extract local feature representation. The DCNNFE block employs the five-layer convolutional backbone of

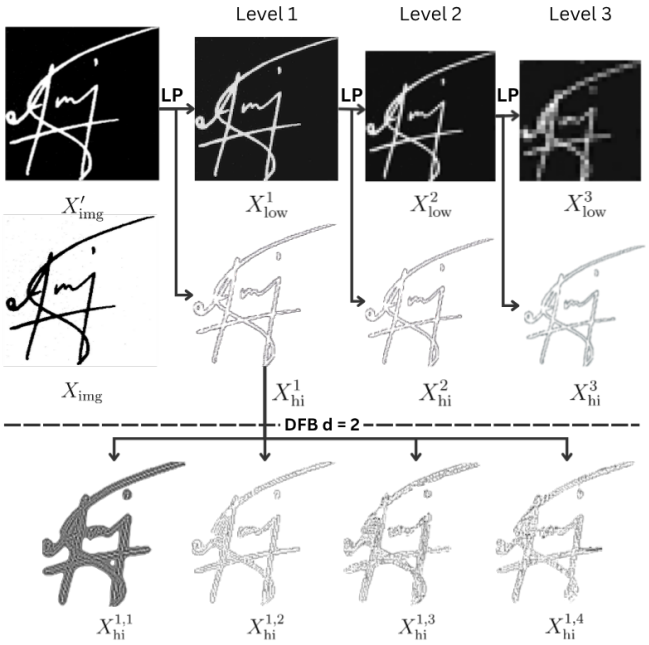


FIGURE 2. Multi-scale and multi-directional decomposition of a signature image using the Contourlet Transform. The Laplacian Pyramid (LP) extracts low-frequency structural information, while the Directional Filter Bank (DFB) produces high-frequency directional subbands.

SigNet [7], replacing its original two fully-connected layers with adaptive global average pooling followed by a single linear projection; the full layer specifications are listed in Table 1. Specifically, the first convolutional layer uses an 11×11 kernel with stride 4 to aggressively downsample the input, followed by successive 5×5 and 3×3 convolutions with increasing channel depths ($96\rightarrow256\rightarrow384\rightarrow384\rightarrow256$). Each convolutional stage is followed by Batch Normalization and a ReLU activation, with three MaxPool layers (kernel 3, stride 2) interleaved to progressively shrink the spatial resolution. The resulting 256-channel feature map is then summarised by adaptive global average pooling and projected to the d-dimensional embedding via a single linear layer, producing $\mathrm{F_{DF}}$. This replaces the original two large fully-connected layers ($6400\rightarrow2048\rightarrow\mathrm{d}$), reducing DCNNFE from 16.97M to 3.74M parameters while preserving the full convolutional capacity and improving generalisation through reduced overfitting risk. Batch Normalization is preferred over Instance Normalization in this branch because DCNNFE operates on reconstructed images whose statistics are more consistent across a mini-batch, and BN is standard in deep CNN backbones [7]. In the OHSV task, using only geometric features extracted by the GFE module makes it harder to capture the local structural features. The DCNNFE module enhances the model's ability to learn local structural features, thereby improving its generalization capabilities. Mathematically, it can be represented as:

$$\mathrm{F_{DF}} = \mathrm{DCNNFE}(\mathrm{X'_{rec}}) \tag{9}$$

where $\mathrm{DCNNFE}(\cdot)$ represents the deep CNN feature extraction module.

E. DEEP SEMANTIC FEATURE EXTRACTION

While geometric features capture high-frequency detail representation and CNNs capture local structural features, the network still lacks global semantic representation, which is important for its generalization capabilities. Given a preprocessed signature image $\mathrm{X'_{img}}$, we extract its semantic features using the DSFE module. The DSFE module consists of two Restormer-style efficient transformer blocks [33], each with 4 attention heads, preceded by CNN stages with ReLU and InstanceNorm. CNN features help the transformer block to learn better [15], while ReLU and InstanceNorm improve gradient flow. MaxPool is used to reduce spatial redundancy and emphasize the most salient features. The CNN stem consists of a point-wise (1×1) convolution followed by a 3×3 convolution that together lift the single-channel input to 32 channels at full resolution. Two downsampling blocks then progressively expand the channel dimension ($32\rightarrow64\rightarrow96$) while halving the spatial resolution through MaxPool ($\mathrm{k}=2$). Each downsampling block uses a 3×3 convolution ($\mathrm{k}=3$, $\mathrm{p}=1$) to preserve local spatial context during channel expansion, followed by a Restormer-style transformer block to capture long-range dependencies at the corresponding scale. Each transformer block is composed of a Multi-head Depth-wise Transposed Attention (MDTA) sub-layer and a Gated Depth-wise Feed-Forward Network (GDFN) sub-layer, both wrapped with pre-LayerNorm and residual connections as proposed in Restormer [33]. Finally, an adaptive global average pooling followed by a linear projection maps the 96-channel spatial response to the d-dimensional semantic feature vector $\mathrm{F_{DS}}$ shared with the other branches. The full layer specifications are given in Table 1.

To capture cross-dependency between geometric and semantic features, we introduce a Cross-Feature Refinement (CFR) block based on bi-directional cross-attention, inspired by task-aware cross-modal feature refinement [34]. Both feature vectors are first projected to a common dimension and reshaped into multi-head subspaces. Each modality then attends to the other via scaled dot-product attention, enabling each feature subspace to selectively incorporate complementary information from the other modality. A learned sigmoid gate adaptively controls the fusion ratio:

$$\begin{aligned} \mathrm{F_{DS}} &= \mathrm{DSFE}(\mathrm{X'_{img}}) \\ \mathrm{g'} &= \mathrm{g} + \mathrm{CrossAttn}(\mathrm{Q{=}g},\ \mathrm{KV{=}s}) \\ \mathrm{s'} &= \mathrm{s} + \mathrm{CrossAttn}(\mathrm{Q{=}s},\ \mathrm{KV{=}g}) \\ \mathrm{F_{SF}} &= \mathrm{LN}(\alpha\cdot\mathrm{g'} + (1-\alpha)\cdot\mathrm{s'}) \end{aligned} \tag{10}$$

where $g = W_g F_{lh}$, $s = W_s F_{DS}$ are learned linear projections, and $\alpha = \sigma(W_\alpha[g', s'])$ is a gating vector produced by a sigmoid-activated linear layer. DSFE$(\cdot)$ represents the deep semantic feature extractor.

F. FEATURE FUSION

After obtaining the outputs of the semantic features F_{SF}, the deep features F_{DF}, and the contourlet statistical features F_{SCT}, we concatenate them along the channel dimension to produce unified feature embeddings:

$$F_{cat} = \text{Concat}(F_{SF}, F_{DF}, F_{SCT}) \tag{11}$$

Then our DWFFF block with residual connection is used to produce fused cross-modality features. The DWFFF is realized by two MLP layers separated by a 1×1 depth-wise convolution and a ReLU activation function. Therefore the DWFFF Module is formulated as:

$$F_{fused} = \text{DWFFF}(F_{cat}) + F_{cat} \tag{12}$$

where the second term denotes a pre-activation residual identity shortcut that bypasses the depth-wise main path; the two MLPs that the prose refers to are both contained inside the DWFFF$(\cdot)$ operator and are therefore not double-counted.

Here F_{fused} denotes the output of the DWFFF, representing the fused features from the geometric, semantic, and statistical feature extractors. Table 1 provides the complete layer-by-layer configuration of all five modules to support reproducibility. The complete model contains approximately 4.13M trainable parameters in total, of which 3.74M reside in the DCNNFE convolutional backbone and the remaining 0.40M are distributed across GFE, DSFE, CFR, and DWFFF. This is substantially more compact than comparable hybrid CNN–Transformer systems, enabling practical deployment without sacrificing discriminative capacity.

IV. EXPERIMENTAL PROTOCOL

The experimental protocol designed to evaluate the proposed approach is discussed in this section. This section discusses the details of datasets, hyperparameters, and evaluation metrics.

A. DATASETS

We evaluated the proposed OHSV system on four publicly available datasets containing both genuine and forged signatures: CEDAR [35], BHSig260 Hindi [36], BHSig260 Bengali [36], and UTSig [37], as summarized in Table 2. The CEDAR (English) dataset comprises 1,320 genuine and 1,320 forged signatures from 55 writers. The BHSig260 dataset comprises 6,240 genuine and 7,800 forged samples from 260 writers in Hindi and Bengali scripts, with each participant contributing 24 genuine and 30 forged signatures. The UTSig dataset (Persian) comprises 8,280 signatures from 115 writers (27 genuine and 45 forged per writer).

For the Siamese-based model, paired inputs of genuine and forged signatures are required, categorized into positive and negative pairs. Positive pairs consist of two genuine signatures from the same writer, while negative pairs include one genuine and one forged signature. For instance, in the BHSig-B dataset, each writer with 24 genuine and 30 forged samples yields $C_{24}^2 = 276$ genuine–genuine (positive) pairs and $24 \times 30 = 720$ genuine–forged (negative) pairs. To balance the dataset, 276 negative pairs are randomly selected from the available 720. The same partitioning strategy is consistently applied across all datasets, aligning closely with their standard evaluation protocols while ensuring uniformity and comparability across experiments [12]. Across all four datasets, the "forged" samples used to form genuine–forged (negative) pairs are the skilled forgeries supplied with each corpus; no random forgeries (i.e., genuine signatures of unrelated writers used as negatives) are included in either training or evaluation. For UTSig, we adopt the official writer-disjoint training/testing split released by Soleimani et al. [37], with no additional repartitioning beyond pair construction.

B. SYNTHETIC DATASET

We evaluated the robustness of GeoCT-SigNet against synthetic signatures produced by two deep generative models: a Variational Autoencoder (VAE) [22] and a Conditional Generative Adversarial Network (CGAN) [21], following the implementation protocol of Ngo et al. [20].

VAE. The VAE encoder maps each input signature image to a lower-dimensional latent space modelled as a Gaussian distribution; the decoder reconstructs a signature from a sampled latent vector. Training minimises the combined objective

$$\mathcal{L}(\theta, \phi; x) = -\mathbb{E}[\log p_\theta(x|z)] + \text{KL}(q_\phi(z|x) \,\|\, p(z)), \tag{13}$$

where the first term is the reconstruction loss and the second is the Kullback–Leibler divergence regulariser [20]. Each VAE was trained for 800 epochs and nine synthetic signatures were produced per reference image. The KL-divergence term provides inherent regularisation over the latent space, enabling VAE to generate outputs that closely reproduce the structural appearance of the reference signatures, as reflected by the high SSIM scores in Table 3..

CGAN. The CGAN extends the vanilla GAN by conditioning both the generator G and discriminator D on a binary class label $y \in \{0, 1\}$ (genuine or forged) [21], optimising

$$\min_G \max_D V(D, G) = \mathbb{E}[\log D(x|y)] + \mathbb{E}[\log(1 - D(G(z|y)))]. \tag{14}$$

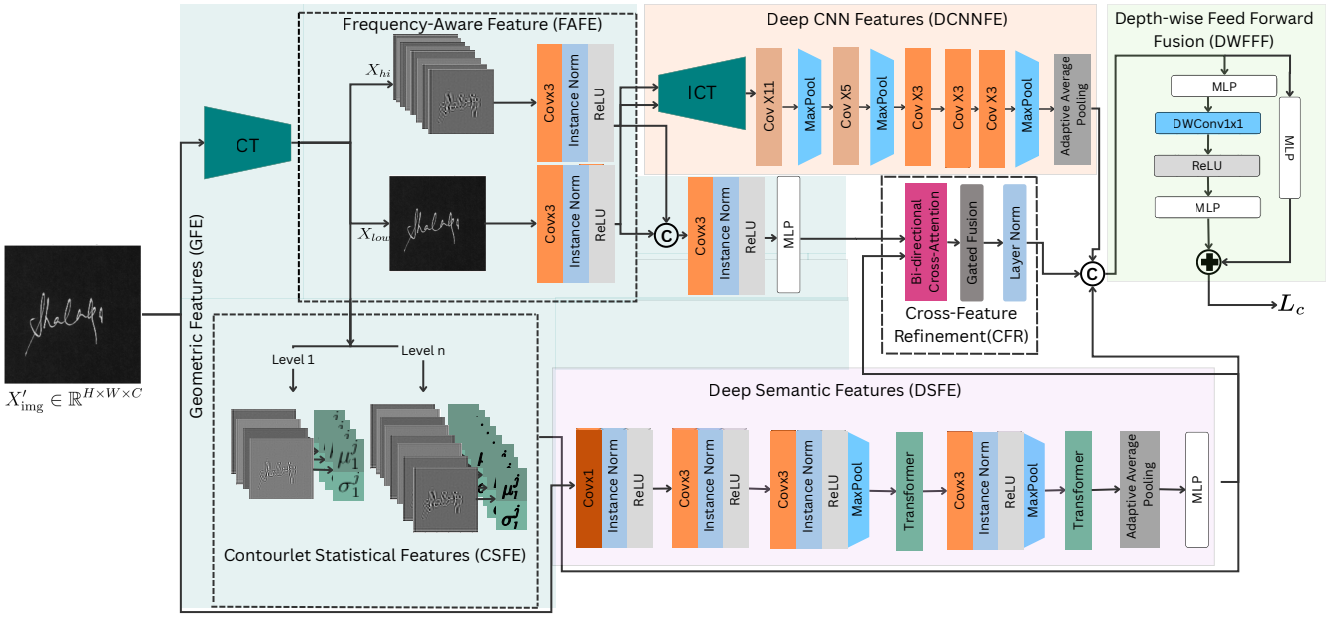


FIGURE 3. The proposed GeoCT-SigNet combines contourlet-based geometric features, CNN-based local structural features, and transformer-based semantic features. These representations are fused through a depth-wise feed-forward module to generate the final embedding for contrastive loss. In every block abbreviation used in this figure and throughout the paper, the trailing letter "E" denotes *Extraction* (i.e., GFE: Geometric Feature Extraction; FAFE: Frequency-Aware Feature Extraction; DCNNFE: Deep CNN Feature Extraction; DSFE: Deep Semantic Feature Extraction). CFR denotes Cross-Feature Refinement and DWFFF denotes Depth-wise Feed-Forward Fusion. The box labelled CSFE (Contourlet Statistical Feature Extraction) is a parameter-free sub-component *within* GFE; it computes the per-subband mean and standard deviation of the contourlet coefficients and carries no learnable parameters, as detailed in Section III-C1 and Table 1.

Individual CGAN models were trained per dataset following Ngo et al. [20].

Whereas Ngo et al. [20] used impostor (forgery) samples as the generation reference, we instead used genuine signature samples for both the VAE and CGAN. This modification follows the threat model of Bird et al. [19], which simulates a realistic scenario in which an imposter has acquired a small number of the target writer's authentic signatures and exploits a generative model to produce synthetic duplicates; a forgery detection system that cannot resist this attack provides weaker security guarantees than one that can.

In all cases, the number of synthetic signatures matched the size of the genuine test sets for CEDAR, BHSig260-Bengali, BHSig260-Hindi, and UTSig. All synthetic datasets were reserved exclusively for testing to prevent data leakage during training. SSIM scores of the generated signatures are reported in Table 3.

C. PREPROCESSING

Signature images are preprocessed by resizing to 224×224 using bicubic interpolation, inverting to a black background with white foreground, and normalizing by the dataset's pixel standard deviation. Fig. 4 shows examples of a genuine signature, an expert forgery, and synthetic signatures.

D. GEOMETRIC FEATURE EXTRACTION HYPERPARAMETERS

We employ the Contourlet Transform (CT) as discussed in Section III. The CT decomposes the input image into multi-scale and multi-directional subbands using the Laplacian Pyramid (LP) for scale decomposition and the Directional Filter Bank (DFB) for directional analysis. Given an input image, we set the decomposition level of the LP to $L = 2$ using the maxflat filter, resulting in two scales of sizes 112×112 and 56×56. The DFB decomposition depth at each LP scale is set to $d = 3$ using the dmaxflat7 filter, yielding $2^d = 2^3 = 8$ directional subbands per scale. The ablation study in Section V demonstrates the effectiveness of these hyperparameter settings.

E. MODEL HYPERPARAMETERS

The model was trained using the Adam optimizer with a learning rate of 1×10^{-3}. A contrastive loss function with a margin of 1.0 was applied, leveraging Euclidean distance. The model was trained for 50 epochs using an Nvidia A6000 GPU. The mini-batch size was set to 32 signature pairs. A step-decay learning-rate schedule was used together with weight decay; both contributed to stable convergence within the 50-epoch budget. The remaining Adam hyperparameters ($\beta_1, \beta_2, \epsilon$) were left at their default values. For every dataset, we adopted the standardized writer-disjoint split established by the

TABLE 1. Complete layer-by-layer configuration of GeoCT-SigNet. Abbreviations: IN = Instance Normalisation, BN = Batch Normalisation, DW = Depth-wise, MHA = Multi-Head Attention, LN = Layer Norm, σ = Sigmoid, k = kernel size, p = padding.

Module	Layer	Operation	Configuration
GFE	CT decompose	Contourlet Transform	2 LP level; DFB depth = 3
	Low-freq. conv	Conv2d → IN → ReLU	k = 3, p = 1
	High-freq. conv	Conv2d → IN → ReLU	k = 3, p = 1
	Fusion conv	Conv2d → IN → ReLU	k = 3, p = 1; concat low + high
	Projection	AdaptAvgPool2d(1) + Linear	output F_{lh}
	F_{SCT} stats	μ_n, σ_n per subband	output F_{SCT}
DCNNFE	Conv1	Conv2d → BN → ReLU → MaxPool	k = 11
	Conv2	Conv2d → BN → ReLU → MaxPool	k = 5, p = 2
	Conv3	Conv2d → BN → ReLU	k = 3, p = 1
	Conv4	Conv2d → BN → ReLU	k = 3, p = 1
	Conv5	Conv2d → BN → ReLU → MaxPool	k = 3, p = 1
	Projection	AdaptAvgPool2d(1) + Linear	output F_{DF}
DSFE	Stage 1a	Conv2d → IN → ReLU	k = 1
	Stage 1b	Conv2d → IN → ReLU	k = 3, p = 1
	Downsample 1	Conv2d → IN → ReLU → MaxPool	k = 3, p = 1
	TransBlock 1	MDTA + GDFN [33]	4 heads
	Downsample 2	Conv2d → IN → ReLU → MaxPool	k = 3, p = 1
	TransBlock 2	MDTA + GDFN [33]	4 heads
	Projection	AdaptAvgPool2d(1) + Linear	output F_{DS}
CFR	Input projection	Linear × 2	project F_{lh}, F_{DS}
	Bi-dir. cross-attn	Scaled dot-product MHA [34]	4 heads; geo↔sem
	Gated fusion	Linear → σ	$\alpha \cdot g' + (1-\alpha) \cdot s'$
	Output	Linear + LN	output F_{SF}
DWFFF	Concat input	[F_{SF}, F_{DF}, F_{SCT}]	concat
	Main path	Linear → DW-Conv1d → ReLU → Linear	DW-Conv1d k = 1
	Residual	Linear	skip connection
	Output	Element-wise add	output F_{fused}

TABLE 2. Dataset Details Including Writers for Training/Testing and Positive/Negative Pairs Per Writer. The "Writers for Testing (Synthetic)" column refers to the same individuals listed under "Writers for Testing (Handwritten)"; synthetic forgeries are generated from those test writers' genuine signatures and used only at test time, so the synthetic and handwritten test pools are not disjoint sets of writers.

Dataset	Writers for Training (Handwritten)	Writers for Testing (Handwritten)	Writers for Testing (Synthetic)	Positive/Negative Pairs per Writer (P/N)
CEDAR	50	5	5	276/276
BHSig-H	100	60	60	276/276
BHSig-B	50	50	50	276/276
UTSig	60	55	55	351/351

TABLE 3. Comparison of VAE and CGAN SSIM Scores for Different Datasets

Dataset	VAE SSIM Score	CGAN SSIM Score
CEDAR	0.504	0.123
BHSig-B	0.922	0.104
BHSig-H	0.916	0.108
UTSig	0.932	0.150

original benchmark papers and used by all baselines in Table 4 to ensure that our reported numbers are directly comparable. For the cross-dataset evaluation in Table 13, a single trained checkpoint per training dataset was reused across the four test domains without per-domain re-tuning.

F. EVALUATION METRICS

Evaluation of the proposed method is measured through four metrics: False Acceptance Rate (FAR), False Rejection Rate (FRR), Equal Error Rate (EER), and Accuracy (ACC). FAR quantifies the system's tendency to accept non-genuine signatures as genuine; FRR quan-

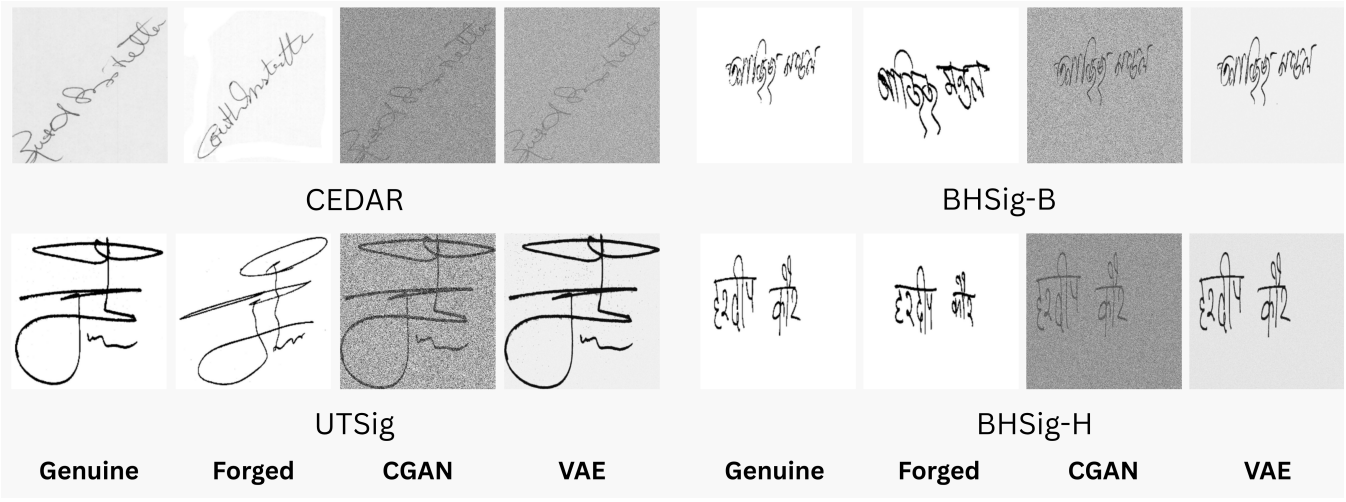


FIGURE 4. Sample of a genuine handwritten signature, expert forgery, and synthetic signatures generated by VAE and CGAN of four datasets.

tifies the system's tendency to reject genuine signatures; and the EER is defined as the error rate when FAR ≈ FRR. The EER is determined by adjusting the threshold until both rates become approximately equal.

V. RESULTS AND DISCUSSION

This section presents the experimental results and analysis of the proposed method, including evaluations against expert forgeries and synthetic signatures, comparisons with state-of-the-art (SOTA) methods, ablation studies on component contributions, sensitivity analysis of key hyperparameters, assessment of preprocessing effects, and cross-dataset validation for generalization. All results are reported in %, with the best results highlighted in bold.

A. EVALUATION OF THE PROPOSED METHOD

We have evaluated the performance of the proposed method in two distinct scenarios against expert forgery and synthetic signatures. For both scenarios, we have followed an established training strategy by training the system with genuine and forged signatures to ensure that no synthetic data leakage occurs during training.

1) Comparison With SOTA Systems on Expert Forgery Signatures

In the first scenario, we have evaluated the proposed method across four datasets and compared it with other SOTA methods. As shown in Table 4, GeoCT-SigNet attains state-of-the-art performance on three of the four benchmarks (BHSig-H, BHSig-B, and CEDAR) and achieves the best overall performance on UTSig in terms of EER (18.99%) and accuracy (81.01%). While DMML reports a lower FAR (9.09%), it does so at the cost of a 32.42% FRR—rejecting roughly one in three genuine signatures. Similarly, GED achieves a lower FRR (7.88%) but incurs a 29.49% FAR. In contrast, GeoCT-SigNet maintains a well-balanced trade-off between FAR and FRR, making it the most reliable method on this benchmark.

GeoCT-SigNet achieves 0% EER (100% accuracy) on CEDAR, a result also reported by SigNet [7], 2C2S [14], 2C2L [25], and PAST [12]. Xiong et al. [12] attribute this to a background bias inherent in the dataset: CEDAR signatures retain writer-specific paper background from the scanning process, enabling models to reach perfect accuracy by differentiating background textures rather than genuine signature strokes. Since this ceiling is shared by multiple methods, the primary quantitative comparisons in this paper are drawn from BHSig-B, BHSig-H, and the synthetic-forgery evaluation in Section V.

2) Comparison With SOTA Systems on Synthetic Signatures

In the second scenario, we evaluate the proposed OHSV system against synthetic duplicates of genuine signatures generated using two popular models: CGAN and VAE. This simulates an imposter who uses a few genuine samples to produce duplicates via generative models. The proposed method consistently achieves the lowest FAR for both VAE and CGAN generated signatures. To benchmark robustness against synthetic forgeries, we compare GeoCT-SigNet against three categories of baselines: (i) popular CNN backbones, namely VGG-16 [10], DenseNet [8], and ResNet [9]; (ii) a Swin Transformer backbone [46]; and (iii) two state-of-the-art OHSV methods with publicly available code, SigNet [7] and HTCSigNet [16]. For consistency, the backbone models in categories (i) and (ii) [8]–[10], [46] are trained and evaluated within the same Siamese framework and under the identical preprocessing pipeline used by our method, whereas the state-of-the-art methods [7], [16] follow their original implementations to ensure a faithful and unbiased evaluation. All synthetic-forgery FAR

TABLE 4. Comparison of Various OHSV Systems on Four Signature Datasets

Dataset	Model	FAR	FRR	ACC	EER
BHSig-H	SigNet [7]*	15.36	15.36	84.64	15.36
	MSN [26]*	17.06	5.16	88.88	11.31
	LBP and ULBP [36]*	24.47	24.47	75.53	24.47
	2C2S [14]*	8.66	9.98	90.68	9.32
	2C2L [25]*	–	–	86.66	13.34
	SURDS [38]*	12.01	8.98	89.50	–
	IDN [27]*	8.99	4.93	93.04	–
	CNN-CrossViT [17]*	8.12	4.61	92.33	7.67
	GeoCT-SigNet	6.89	6.80	93.16	6.84
BHSig-B	SigNet [7]*	13.89	13.89	86.11	13.89
	MSN [26]*	10.42	6.44	91.56	8.43
	LBP and ULBP [36]*	33.82	33.82	66.18	33.82
	TransOSV [13]*	9.90	9.90	–	9.90
	HTCSigNet [16]*	9.20	7.83	91.17	8.52
	2C2S [14]*	5.37	8.11	93.25	6.75
	2C2L [25]*	10.44	9.37	88.08	11.92
	SURDS [38]*	19.89	5.42	87.34	–
	LBPKNN [39]*	–	–	90.36	–
	AVN [40]*	–	–	93.80	6.14
	GeoCT-SigNet	5.85	5.74	94.20	5.80
CEDAR	SigNet [7]*	0	0	100	0
	MSN [26]*	3.18	0	98.40	1.63
	2C2S [14]*	0	0	100	0
	2C2L [25]*	–	–	100	0
	LQP [41]*	5.01	6.12	–	–
	BFS [42]*	4.67	4.67	–	–
	PAST [12]*	0	0	100	0
	GeoCT-SigNet	0	0	100	0
UTSig	UTSig [37]*	21.29	39.27	–	–
	DMML [43]*	9.09	32.42	–	–
	FCN [44]*	32.43	32.50	–	–
	GED [45]*	29.49	7.88	–	–
	PAST [12]*	21.08	21.18	78.87	21.13
	GeoCT-SigNet	19.05	18.93	81.01	18.99

* Indicates all values taken from the original paper of the respective method.

values reported in Table 5 are computed at the same operating threshold as the corresponding real-forgery EER point in Table 4; we deliberately do not fine-tune the threshold on synthetic samples, so the reported numbers reflect out-of-the-box robustness. The same convention is applied to every baseline in the table. Our results indicate that CNN-based OHSV systems exhibit high vulnerability, with FAR values reaching 60.87% on BHSig-H (using the ResNet backbone) and 47.46% on BHSig-B for VAE generated signatures. CGAN duplicates yield FARs of up to 86.7% on CEDAR and 52.43% on BHSig-B. Higher SSIM (Table 3) means the synthetic is closer in appearance to the genuine reference, so VAE duplicates (SSIM 0.92–0.93 on BHSig and UTSig) reproduce the stroke geometry that CNN baselines rely on and push their FAR above 60% [20]. CGAN SSIM is low (0.10–0.15) because the discriminator, trained on limited per-class samples and conditioned only on a genuine/forged label, learns class-level statistics rather than writer-specific geometry [19]; nevertheless such outputs can still fool CNN baselines (ResNet reaches 86.7% FAR on CEDAR-CGAN) by drifting outside the learned forgery distribution [20]. GeoCT-SigNet holds 0% FAR on CGAN and $\leq$ 3.6% on VAE across all four datasets because its contourlet descriptors capture multi-directional stroke structure and flag the distortions these generators leave behind, regardless of SSIM. Transformer-based models, such as Swin Transformer, perform more robustly due to their improved long-range spatial modeling capabilities. HTCSigNet [16] is a hybrid transformer–convolution model, yet it degrades the most under synthetic attack. Its expert-forgery FAR is low (4.65% on BHSig-H, 0% on CEDAR), but its FAR reaches 38.50% (CGAN) and 23.80% (VAE) on BHSig-H and 44.73% (CGAN) and 35.80% (VAE) on BHSig-B. We attribute this to its decision head. HTCSigNet separates representation learning from classification and uses a per-writer RBF-SVM trained on genuine and

skilled-forgery samples [16]. This boundary is fitted to human forgeries and does not transfer to out-of-distribution inputs, so synthetic samples beyond the SVM's training support are often accepted as genuine. The failure is largest for CGAN, whose duplicates are the least similar to the genuine references (SSIM 0.10–0.15, Table 3) and therefore lie farthest from the modelled feature region.

Siamese verifiers (SigNet [7], the Swin-transformer [46] backbone, and GeoCT-SigNet) are more robust and keep CGAN-FAR low. A Siamese network does not learn a fixed class boundary; it learns an embedding under a contrastive loss and decides from the distance between a query and a genuine reference. Because the decision is reference-relative, it does not assume that test forgeries match the training distribution.. The property that makes CGAN duplicates hard for an SVM head, their distance from genuine signatures, is precisely what makes them easy for a metric-based verifier to reject. GeoCT-SigNet benefits further from its contourlet descriptors, which capture the multi-directional stroke distortions of synthetic signatures and reduce CGAN-FAR to 0% on all four datasets. Results are summarized in Table 5.

B. ABLATION STUDIES

In this section, we provide ablation studies to demonstrate our contributions. For all experiments in this section, we used the BHSig-B dataset. We have followed the same experimental protocols as mentioned in Section IV. We have provided ablation on proposed feature extraction blocks, hyperparameter sensitivity, and fusion strategy.

1) Ablation of GeoCT-SigNet Components

We have assessed the impact of each feature extraction block in our proposed network. As shown in Table 6, when we used the DFE (i.e., CNN) block alone, our model achieved an EER of 11.08%. We observed that using a frequency-refined image in the DFE block (i.e., FAFE + DFE) resulted in a drop in our model's performance, because the CNN model has limited receptive fields, leading to the loss of the frequency-refined information introduced by FAFE. In comparison, SFE (i.e., transformer block) achieves better performance with geometrically incorporated (GFE) representation (EER 8.34%). Other than that, when we use combined transformer and CNN (i.e., DFE + SFE) blocks, we achieved EER 7.03%. Finally, integrating geometric information with both CNN and transformer modules (DFE + SFE + GFE) yields the lowest EER of 5.8%, highlighting the effectiveness of combining structural, semantic, and geometric feature representations.

2) Impact of GFE on Synthetic Signatures

We assessed the impact of Geometric Features (GF) on synthetic signatures generated by generative models. As shown in Table 7, GF significantly improves the performance of the proposed method against VAE and CGAN signatures. This improvement demonstrates that the GF representation effectively captures meticulous high-frequency details (such as strokes and edges), thereby significantly improving the performance of the proposed method against synthetic signatures.

3) Feature Map Visualization

To qualitatively illustrate the effect of the GFE module, we applied Gradient-weighted Class Activation Mapping (Grad-CAM) [47] to four model configurations: DCNNFE alone, DCNNFE + GFE, DSFE alone, and DSFE + GFE. Activation maps were extracted from the final convolutional layer of each branch for four signature categories from the BHSig260-Bengali test set: Genuine, Expert Forgery, Synthetic VAE, and Synthetic CGAN. Fig. 5 shows the results alongside the preprocessed input images (top row).

Without GFE, both branches exhibit diffuse activations. The CNN branch (DCNNFE) produces coarse, spread-out hotspots that fail to localise stroke-level differences between genuine and forged inputs, while the Transformer branch (DSFE) traces broad stroke outlines across the full signature. For CGAN inputs, DCNNFE generates a near-uniform rectangular activation driven by noise texture rather than signature content, consistent with the high CGAN-FAR observed in Table 5.

Adding GFE significantly sharpens the activations in both branches. DCNNFE + GFE produces compact hotspots concentrated at high-curvature stroke regions and pen-lift points, with the noise-driven rectangular response for CGAN inputs collapsing to a sparse, localised map. DSFE + GFE retains the fine-grained stroke-boundary attention of the Transformer while further concentrating activations at geometrically distinctive stroke junctions. Collectively, these maps confirm that the GFE module acts as a geometric attention prior, redirecting both branches towards discriminative stroke cues and suppressing noise activations across all forgery types.

4) Contrastive Loss Margin Parameter

We have assessed the impact of the contrastive loss margin parameter. Our results, as shown in Fig. 6, indicate that the margin parameter set to 1.0 yields the best performance (i.e., EER 5.8%). Our results provide a similar analysis to that of previous studies [7], [26], which show that for Siamese-based signature verification systems, a margin of 1.0 tends to yield better performance.

TABLE 5. Comparison of SOTA Systems on Four Datasets. Expert forgery indicates a human was involved in creating the forged signatures. Synthetic indicates signatures generated by CGAN and VAE. FAR indicates % of forged or synthetic signatures accepted as genuine.

Dataset	Feature Extractor	Expert Forgery Attack (FAR)	Synthetic (VAE FAR)	Synthetic (CGAN FAR)
BHSig-H	VGG16 [10]	18.14	23.23	21.71
	DenseNet [8]	25.30	19.57	16.50
	ResNet [9]	31.42	60.87	56.98
	SigNet Siamese [7]	13.24	32.65	15.75
	HTCSigNet [16]	4.65	23.8	38.5
	SWIN Transformer [46]	18.06	19.07	4.20
	GeoCT-SigNet	6.89	2.60	0
BHSig-B	VGG16 [10]	19.10	30.90	16.09
	DenseNet [8]	19.60	14.97	1.60
	ResNet [9]	21.08	47.46	52.43
	SigNet Siamese [7]	11.04	21.57	21.57
	HTCSigNet [16]	9.20	35.80	44.73
	SWIN Transformer [46]	12.02	17.69	1.64
	GeoCT-SigNet	5.85	2.13	0
CEDAR	VGG16 [10]	0.40	12.37	7.40
	DenseNet [8]	2.10	10.40	76.30
	ResNet [9]	12.30	9.60	86.70
	SigNet Siamese [7]	0	15.60	14.60
	HTCSigNet [16]	0	25.30	24.60
	SWIN Transformer [46]	0	0	0
	GeoCT-SigNet	0	0	0
UTSig	VGG16 [10]	34.66	43.08	30.05
	DenseNet [8]	36.09	43.08	30.05
	ResNet [9]	34.15	62.79	5.78
	SigNet Siamese [7]	36.73	26.74	27.53
	HTCSigNet [16]	--	29.52	34.23
	SWIN Transformer [46]	31.30	8.23	0.53
	GeoCT-SigNet	19.05	3.60	0

All models were trained on only human-written genuine and forged signatures and evaluated on synthetic datasets.

TABLE 6. Ablation Study Showing the Impact of Each Proposed Feature Component on System Performance

Deep Features (DF)		Geometric Features (GF)			Metrics			
DFE	SFE	GFE	FAFE	CFR	FAR	FRR	ACC	EER
		✓			23.56	23.41	76.51	23.49
		✓	✓		19.56	19.41	80.51	19.49
✓			✓		16.86	16.75	83.19	16.81
✓		✓	✓		11.24	11.19	88.78	11.22
✓					11.04	11.13	88.92	11.08
	✓	✓			9.38	9.32	90.65	9.30
	✓				8.58	8.62	91.40	8.60
	✓	✓	✓	✓	8.32	8.37	91.66	8.34
✓	✓		✓		7.65	7.69	92.33	7.67
✓	✓				7.17	6.89	92.97	7.03
✓	✓	✓	✓	✓	5.85	5.74	94.20	5.80

5) Impact of Fusion Strategy

We evaluated our fusion strategy against simple concatenation and element-wise sum. Concatenation merges features from all branches, while sum combines key characteristics of input features. Our proposed DWFFF fusion strategy outperforms both, as shown in Table 8. Details are provided in Section III.

6) Contourlet Transform Parameters

We assessed the impact of the contourlet transform parameters for the GFE block. For this, we conducted an ablation study on both the multi-scale (LP) decompositions and multi-directional (DFB) decomposition. For the multi-scale decomposition, we varied the Laplacian Pyramid levels (levels 1 through 4) while keeping the DFB depth fixed at d = 3. ... L = 2 and d = 3 (yielding $2^3 = 8$ directional subbands per scale) provided the best performance.

7) Effect of Geometric Transform Selection

We conducted an ablation study to evaluate the effect of the geometric transform used inside the GFE mod-

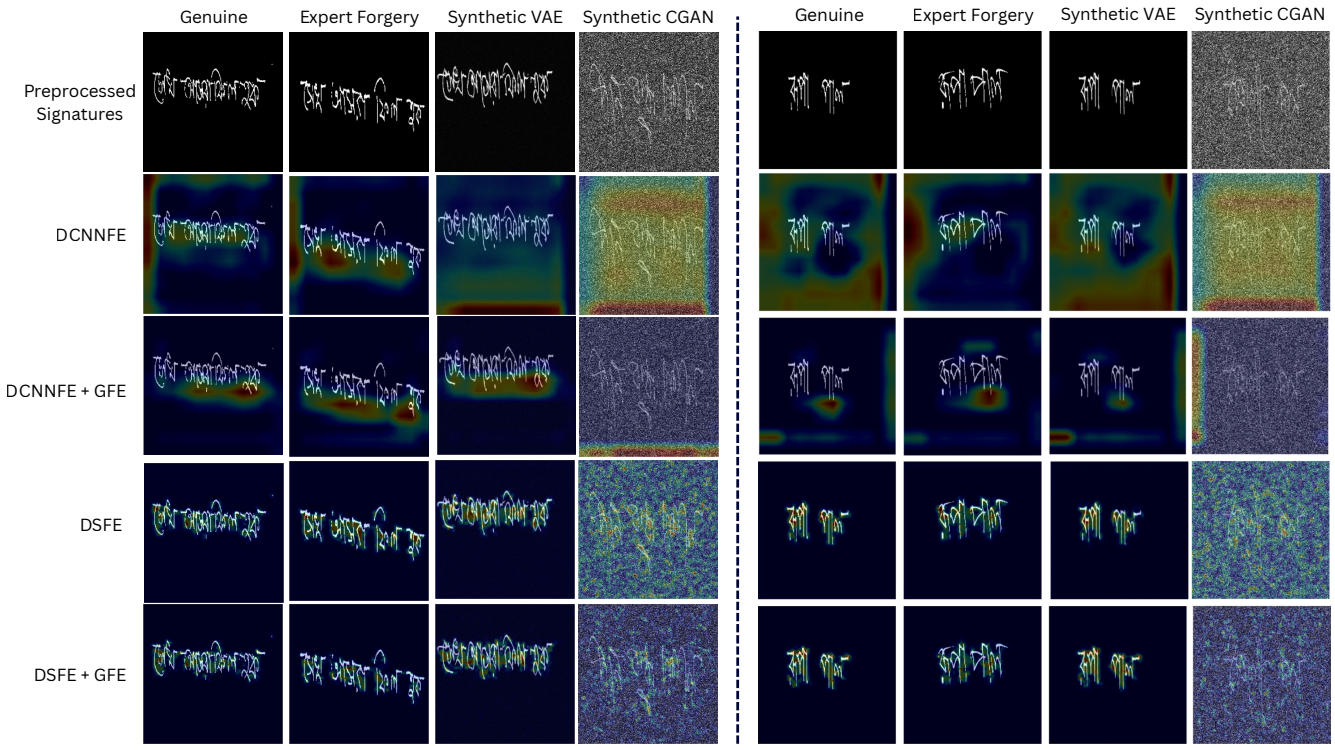


FIGURE 5. Visualization of attention maps on the BHSig260-Bengali dataset. The first row shows the original preprocessed signature images. The second row shows the attention maps overlaid on signatures using the DCNNFE branch without the GFE module. The third row shows the attention maps overlaid using the DCNNFE branch with the GFE module. The fourth row shows the attention maps overlaid using the DSFE branch without the GFE module. The fifth row shows the attention maps overlaid using the DSFE branch with the GFE module. The four columns represent four signature categories: genuine signatures, expert forgeries, synthetic signatures generated by VAE, and synthetic signatures generated by CGAN, respectively.

TABLE 7. Performance of the GF Extraction Block on Expert Forgery and Synthetic Signatures. CSF Extraction Block Inclusion Significantly Improves Performance on Synthetic Signatures.

Features Extractor	Expert Forgery (ACC)	Synthetic VAE (ACC)	Synthetic CGAN (ACC)
with CSF	94.20	97.87	100.00
w/o CSF	94.16	84.26	99.27
w/o GFE	92.97	81.26	97.27

Note: $*$ indicates inclusion of DF and GF blocks unless otherwise specified. Moreover, w/o means without.

TABLE 8. Performance Comparison of Different Feature Fusion Strategies

Fusion	FAR	FRR	ACC	EER
Sum	7.30	7.15	92.77	7.23
Concat	7.44	7.44	92.56	7.44
DWFFF	5.85	5.74	94.20	5.80

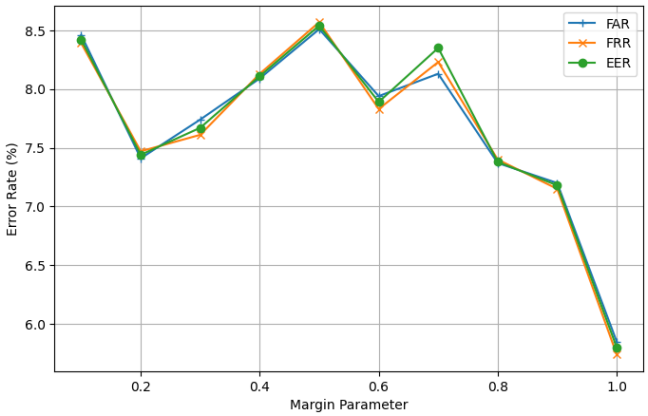


FIGURE 6. Effect of margin parameter on verification performance.

ule. Three representative multi-scale, multi-directional transforms were compared: (i) the Contourlet Transform (CT), which decomposes an image through a Laplacian Pyramid followed by a Directional Filter Bank; (ii) the Discrete Wavelet Transform (DWT) [30] using Haar filters with two decomposition levels; and (iii) a Gabor Filter Bank [48] with four orientations and two scales. All other modules (DCNNFE, DSFE, CFR, DWFFF) remained identical across runs. As shown in Table 10, the Contourlet Transform consistently outperforms both alternatives on expert forgery as well as synthetic signatures generated by VAE and CGAN. The superior performance of the Contourlet Transform can be attributed to its anisotropic directional selectivity and near-critical sampling, which preserve fine stroke-level

TABLE 9. Comparison of Signature Verification Results (Accuracy %) With DFB and LP

Methods	Accuracy	Methods	Accuracy
DFB = 2	92.16	LP = 1	91.16
DFB = 4	93.25	LP = 2	94.20
DFB = 8	94.20	LP = 3	93.40
DFB = 16	93.01	LP = 4	92.70

TABLE 10. Ablation study comparing geometric transforms in the GFE module on BHSig260 Bengali. Accuracy (%) is reported for expert forgery (E.F.), VAE-generated, and CGAN-generated signatures.

Transform	Expert Forgery (ACC)	Synthetic VAE (ACC)	Synthetic CGAN (ACC)
Gabor	92.14	94.00	97.40
Wavelet (Haar DWT)	93.42	95.00	98.00
Contourlet	94.20	97.87	100.00

details more effectively than the isotropic DWT or the fixed-bandwidth Gabor bank.

8) Effect of the Inverse Contourlet Transform (ICT) Reconstruction Step

The FAFE block first refines the low- and high-frequency contourlet sub-bands and then applies an Inverse Contourlet Transform (ICT) to reassemble the refined sub-bands into a geometrically enhanced spatial image, which is forwarded to DCNNFE. To isolate the contribution of this reconstruction step, we compared the full model against a variant in which the ICT was removed and the refined sub-bands were forwarded as a multi-channel feature stack directly into DCNNFE; all other modules (GFE sub-band refinement, DCNNFE, DSFE, CFR, DWFFF) were kept identical. Table 11 reports the BHSig-B (Bengali) expert-forgery results. Including the ICT reconstruction step yields a clear improvement across every metric, indicating that allowing DCNNFE to operate on a spatially reconstructed image makes better use of the network's local convolutional inductive biases.

C. IMPACT OF BACKGROUND PREPROCESSING

This experiment evaluated the impact of background preprocessing on verification performance. Previous studies either performed background removal, applied background inversion to enhance stroke contrast, or did no background processing. We tested three variants: without background (Otsu), with background, and inverse background preprocessing. The inverse background approach performed best, achieving 94.2% accuracy (Table 12), indicating that background inversion enhances stroke contrast and enables the model to capture relevant features.

TABLE 11. Ablation of the Inverse Contourlet Transform (ICT) reconstruction step in FAFE on BHSig-B (Bengali) expert forgery. FAR, FRR, and EER are reported in %; Accuracy (ACC) is reported in %.

Variant	FAR	FRR	ACC	EER
Without ICT (sub-band)	7.65	7.70	92.32	7.68
With ICT (full model)	5.85	5.74	94.20	5.80

TABLE 12. Impact of Background Preprocessing Strategies on Signature Verification Performance

Method	FAR	FRR	ACC	EER
w/o BG	7.27	7.37	92.68	7.32
With BG	7.10	7.05	92.92	7.08
Inverse BG	5.85	5.74	94.20	5.80

D. CROSS-DATASET VALIDATION

We assessed the generalization capability of the proposed method across different languages using four signature datasets: CEDAR, BHSig-B, BHSig-H, and UTSig. As shown in Table 13, the model achieves strong intra-language performance, attaining 100% accuracy on CEDAR, 94.20% on BHSig-B, and 93.16% on BHSig-H. However, performance decreases under cross-language evaluation due to variations in writing styles and scripts. Models trained on BHSig datasets exhibit comparatively better transferability (e.g., 75.18% when trained on BHSig-B and tested on CEDAR), while UTSig shows the lowest cross-language generalization.

E. COMPUTATIONAL COMPLEXITY ANALYSIS

Table 14 compares GeoCT-SigNet against four recent hybrid CNN–Transformer baselines for offline signature verification along two complementary axes: model size (trainable parameters) and inference cost (GFLOPs per forward pass on a 224×224 input pair). At 4.13 M parameters, GeoCT-SigNet is the most compact model in the table by a wide margin: it is 11× smaller than HTCSigNet (94.47 M), 20× smaller than TransOSV (80.91 M), 11× smaller than 2C2S (47.40 M), and 3× smaller than CNN-CrossViT (12.60 M). Three architectural choices account for this compactness. (i) The DCNNFE branch replaces SigNet's two large fully-connected layers ($6400 \to 2048 \to d$) with adaptive global average pooling followed by a single linear projection, reducing the CNN backbone from 16.97 M to 3.74 M parameters while preserving its full convolutional capacity (Section III D). (ii) The DSFE branch uses two Restormer-style efficient transformer blocks with depth-wise attention (MDTA + GDFN) operating on a 96-channel feature map, which is markedly lighter than

TABLE 13. Cross-Dataset Generalization Performance. Each Row Represents the Training Dataset and Each Column the Testing Dataset.

	CEDAR	BHSig-B	BHSig-H	UTSig
CEDAR	100.00	72.94	79.13	56.54
BHSig-B	75.18	94.20	71.12	61.22
BHSig-H	61.19	75.59	93.16	57.21
UTSig	56.99	53.07	55.21	81.01

TABLE 14. Benchmark Computational Complexity Comparison.

Model	Params (M)	GFLOPs
TransOSV [13]*	80.91	33.18
2C2S [14]*	47.40	7.91
HTCSigNet [16]	94.47	48.329
CNN-CrossViT [17]*	12.60	10.55
GeoCT-SigNet (Ours)	4.13	9.599

* Values taken from the authors' papers.

the high-dimensional self-attention used in TransOSV and HTCSigNet. (iii) The contourlet decomposition itself is a fixed non-learnable filter bank (0 trainable parameters) and the FAFE refining convolutions act on small 112×112 and 56×56 sub-bands, contributing only $\sim$ 100 K parameters in total (Table 1). Inference cost reflects the same trade-off: at 9.599 GFLOPs GeoCT-SigNet is below TransOSV (33.18 GFLOPs) and HTCSigNet (48.329 GFLOPs), comparable to CNN-CrossViT (10.55 GFLOPs), and slightly above 2C2S (7.91 GFLOPs) despite carrying an additional geometric branch. Taken together, the parameter and GFLOPs figures indicate a favourable theoretical complexity profile relative to recent hybrid CNN–Transformer baselines.

DATA AVAILABILITY

The four benchmark datasets used in this study (CEDAR, BHSig260 Hindi, BHSig260 Bengali, and UTSig) are publicly available from their respective authors. The source code implementing GeoCT-SigNet, the trained model weights, and the exact writer-disjoint train / validation / test splits used in our experiments will be released at a public GitHub repository upon acceptance, to support reproducibility.

VI. CONCLUSION

This paper introduced a novel geometry-aware hybrid CNN-Transformer signature network (GeoCT-SigNet) for the writer-independent offline signature verification task. GeoCT-SigNet effectively integrates microstructural geometric features with deep semantic features using a depth-wise feedforward fusion block. Extensive experiments across four benchmark datasets show that GeoCT-SigNet matches or surpasses state-of-the-art methods on three of four datasets in detecting expert forgeries and is consistently the most robust system against synthetic forgeries generated by VAE and CGAN at the same operating threshold. Extensive ablation studies underscore the significance of integrating geometric features with semantic features. Together with its compact 4.13 M parameter footprint, these results suggest that contourlet-based geometric priors are a promising direction for parameter-efficient OHSV; we view the present work as motivating future work on deployment-realistic conditions, including multi-script joint training, demographic-stratified evaluation, and resilience against stronger generative-attack families beyond CGAN/VAE.

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